

1. Prove that the trace of the symmetric real matrix is equal to the sum of its eigenvalues (Hint: use the spectral decomposition and the circular property of the trace).
2. Consider the real matrix X of dimension $n \times p$.
 - (a) Prove that $X^\top X$ is semipositive definite and that its eigenvalues are larger or equal to zero.
 - (b) Prove that when $p > n$ then at least one eigenvalue of $X^\top X$ is equal to zero (i.e., $X^\top X$ is singular).
3. In linear regression model the response is distributed as $y \sim \mathcal{N}(X\beta, \sigma^2 I)$. Assuming $\sigma^2 = 1$, write the density (likelihood) $f(y; \beta)$ of the random vector y . Derive $f(y; \hat{\beta}_{MLE})$, where $\hat{\beta}_{MLE}$ is the maximum-likelihood estimator of β , to derive

$$\begin{aligned} AIC &= -2\log(f(y; \hat{\beta}_{MLE})) + 2k \\ BIC &= -2\log(f(y; \hat{\beta}_{MLE})) + k\log(n), \end{aligned}$$

where k is the dimension of the model, (length of β).

4. Your data contains 10 variables. You fit 10 regression models including the first variable, the first two variables, etc. The residual sums of squares for these 10 consecutive models are equal to

$$(1731, 730, 49, 38.9, 32, 29, 28.5, 27.8, 27.6, 26.6).$$

The sample size is equal to 100. Which of these 10 models will be selected by AIC? Which model will be selected by BIC? Assume that the standard deviation of the error term is known; $\sigma = 1$.

5. Assuming the orthogonal design ($X^\top X = I$) and $n = p = 10000$, calculate the expected number of false discoveries for AIC and BIC, when none of the variables is really important (i.e., $\beta_i = 0$ for all i).
6. When would you use AIC vs. BIC?
7. Derive the formula for the bias, variance, and MSE of the ridge regression estimate under the orthogonal design (i.e., when $X^\top X = I$). Compare to the respective values for the least square estimator. Discuss the bias-variance tradeoff for different γ . Is there always a value $\gamma > 0$ such that MSE for ridge is strictly smaller than that of the least squares?
8. For a given data set with 40 explanatory variables, the residual sum of squares from the least squares method and the ridge regression are equal to 4.5 and 11.6, respectively. For the ridge regression, the trace of $X(X^\top X + \gamma I)^{-1}X^\top$ is equal to 32. Which of these two methods yields the better estimated prediction error?
9. Given $X^\top X = I$, calculate the expected value of false discoveries and the power of LASSO.

$$\hat{\beta} = \operatorname{argmin}_{\beta \in \mathbb{R}^p} \frac{1}{2} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1$$

10. Consider adaptive LASSO with $\lambda_i = w_i\lambda$. (for example $w_i = 1/|\hat{\beta}_{OLS,i}|$)

$$\hat{\beta}_{AL} = \underset{\beta \in \mathbb{R}^p}{\operatorname{argmin}} \frac{1}{2} \|y - X\beta\|_2^2 + \sum_{i=1}^p \lambda_i |\beta_i|,$$

- (a) How can you calculate the adaptive LASSO estimator using the numerical solver for LASSO (like `glmnet`)?
- (b) In the orthogonal case ($X^\top X = I$), calculate the value of the adaptive LASSO estimator for the specific coordinate of the beta vector.
- (c) The ordinary least squares estimator of β_1 under the orthogonal design ($X^\top X = I$) is equal to 3 and the LASSO estimator of this parameter is equal to 2. What is the value of the adaptive LASSO estimator of β_1 if we use the same value of λ and the weight for X_1 is $w_1 = 1/4$?